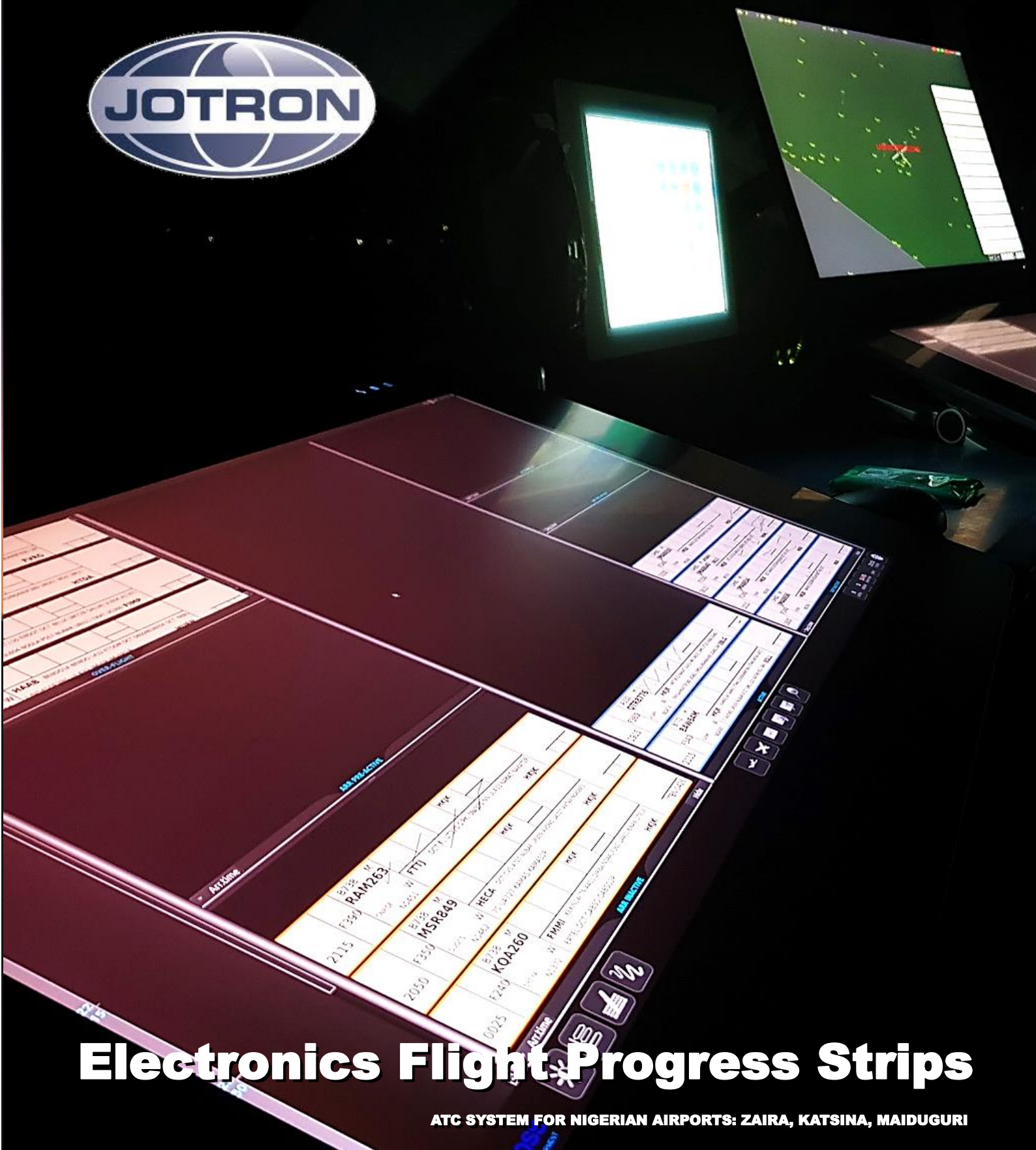




Electronics Flight Progress Strips

ATC SYSTEM FOR NIGERIAN AIRPORTS: ZAIRA, KATSINA, MAIDUGURI

iBross
PROJECT DEVELOPMENT

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Agency



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SYSTEM ADMINISTRATION

jotron.nama.twr3.strips.doc.admin 200313

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1. SYSTEM ARCHITECTURE

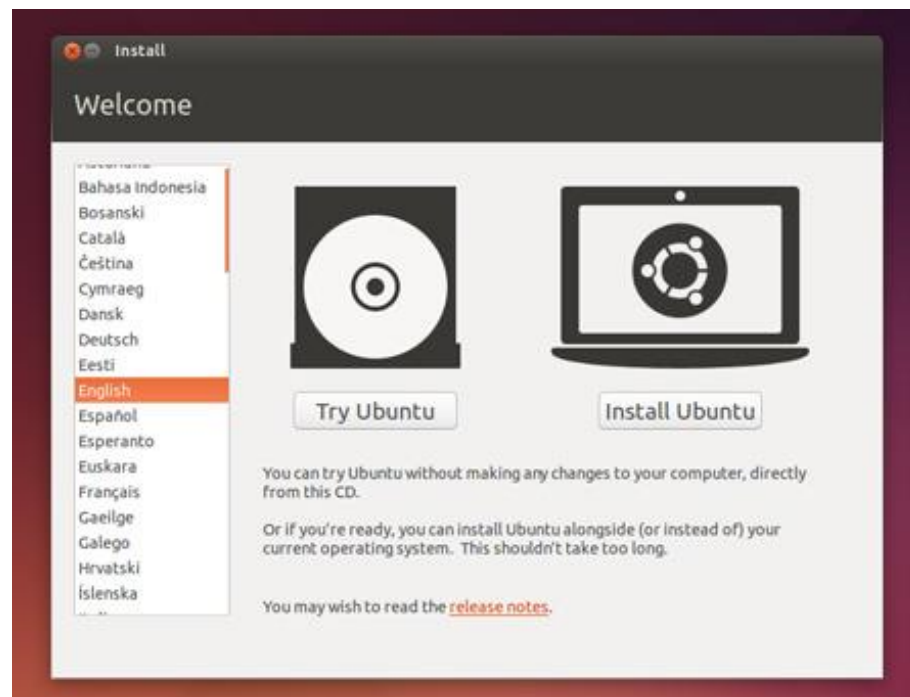
Subject of update after system installed.

2. INSTALLATION

2.1 Operating system

Insert Installation Disk 1/2 (Operational System) into the DVD-drive.

Restart your computer. You should see a welcome screen prompting you to choose your language with the option to install Ubuntu or try it from the DVD.

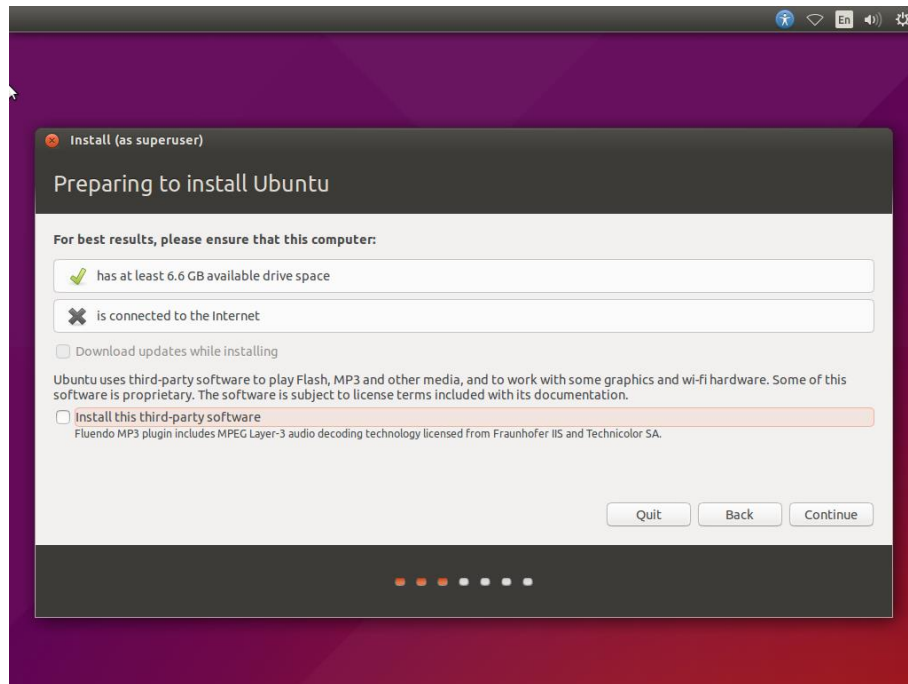


Note: If installation does not start automatically after restarting computer, you should set boot order in BIOS to place boot from DVD-ROM at first place. To do this follow the steps below:

1. Turn on or restart the computer.
2. While the display is blank, press the F10 key to enter the BIOS settings menu.
3. Select the Storage tab using the right and left arrow keys.
4. Use the up and down arrow keys to select Boot Order.
5. Follow the on-screen instructions to change the boot order.

Choose install Ubuntu option.

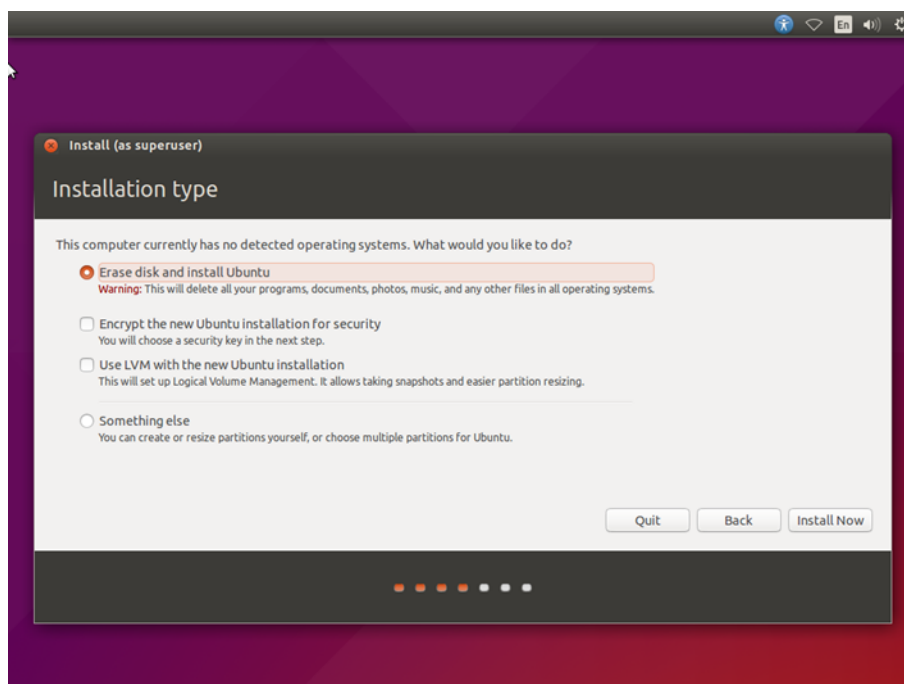
You should see Preparing to install Ubuntu screen.



Note 1: In Ubuntu 18.04.2 order of installation screens are little different.

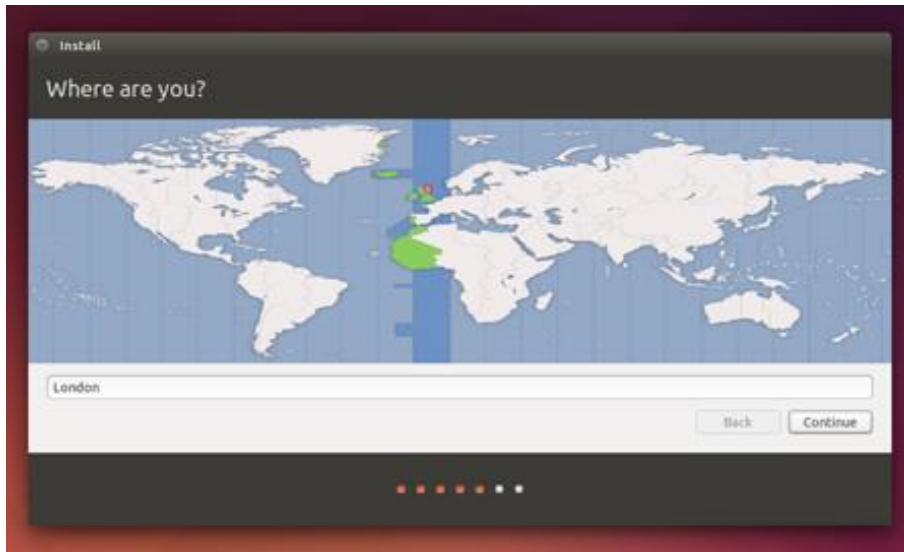
Note 2: In Ubuntu 18.04.2 next screen (**Prepare to install Ubuntu**) is missing, instead of it there is little different screen **Updates and other Software**. You should choose normal installation and then continue to next screen.

Keep all options unchecked and press Continue button.
You should see next screen (Installation type).



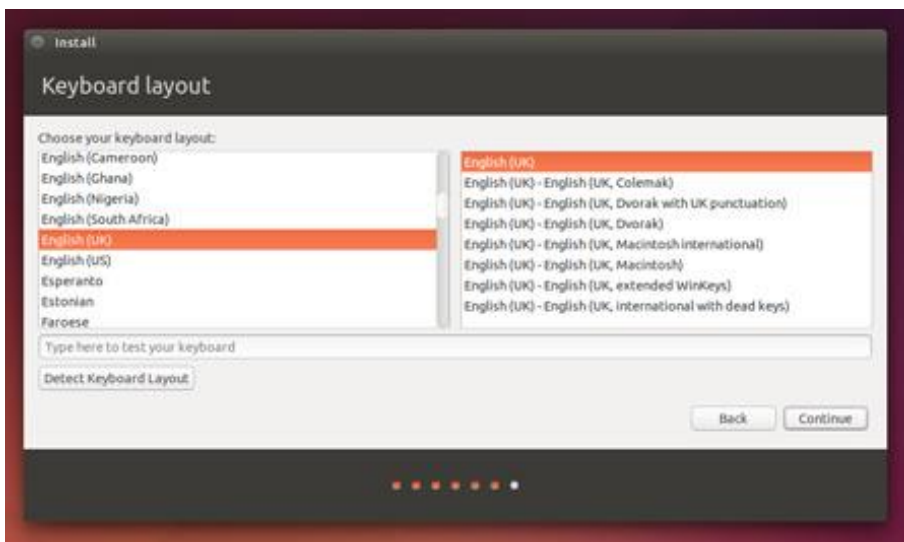
Choose Erase disk and install Ubuntu option and press Install Now button. In the next window (Write the changes to disks?) confirm start of the installation by pressing Continue button.

Then you should see screen for selecting location.



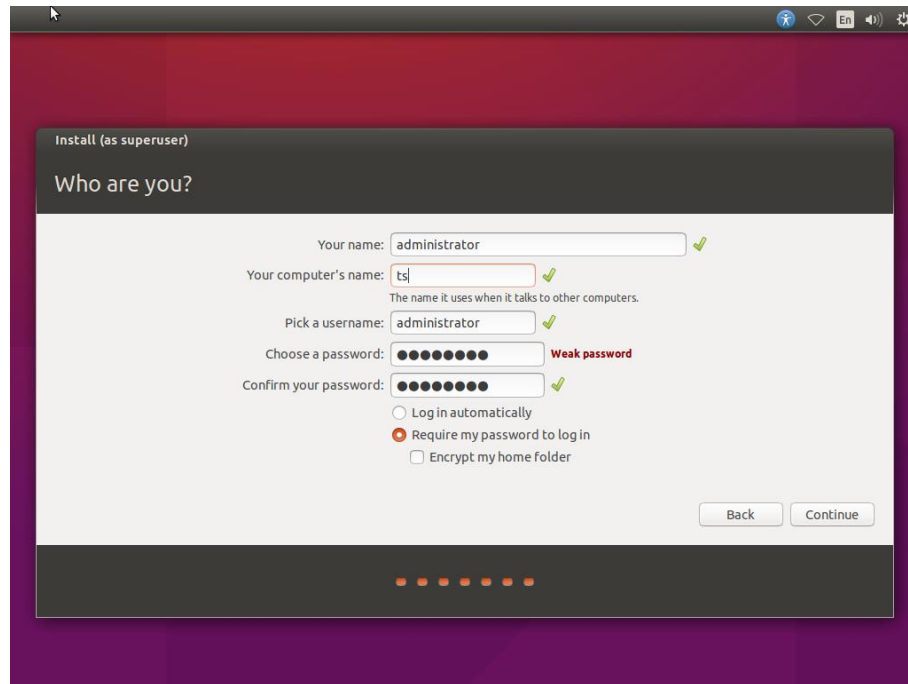
Skipping this step, UTC time zone will be set later, after installation.

Press continue button and Keyboard layout screen should appear.



Choose English (US) options. After selecting layout press Continue button

Then you should see screen for entering login and password details.



Fill appropriate text fields according previous picture. Let Require my password to log in checked.

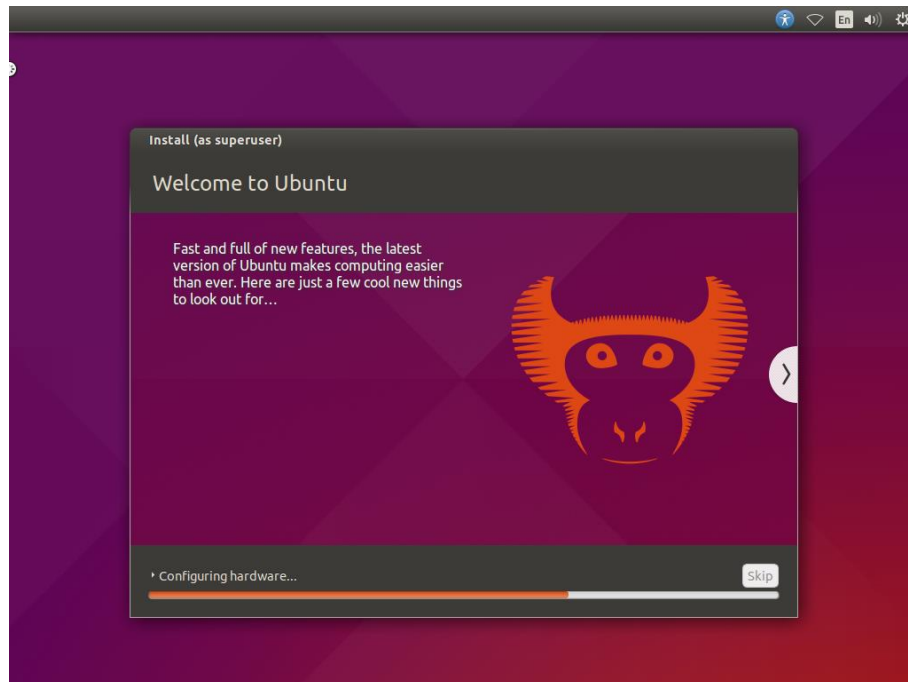
Note 1: This dialog creates only initial user. Another users can be added later, after installation.

Note 2: By default, the root account password is locked in Ubuntu. This means that you cannot login as root directly or use the "su" command to become the root user. However, since the root account physically exists it is still possible to run programs with root-level privileges. This is where sudo comes in - it allows authorized users (normally "Administrative" users) to run certain programs as root without having to know the root password. When sudo asks for a password, it needs your user password.

Note 3: This user is by default "Administrative" user.

After entering login and password details press Continue button.

Next screen should appear which means that installation is in progress.



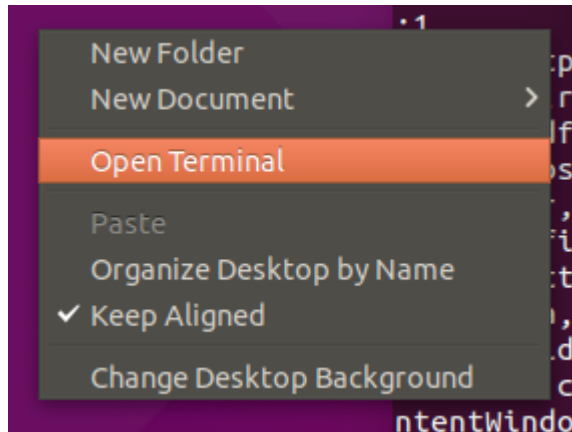
After finishing installation Installation Complete window will appear.

Press "Restart Now" button,,computer will restart.

Installation of operating system is now complete.

2.2 Post operating system installation settings

Log in as "Administrative" user, open terminal application by right mouse click select open terminal in context menu.



2.2.1 Setting UTC time zone

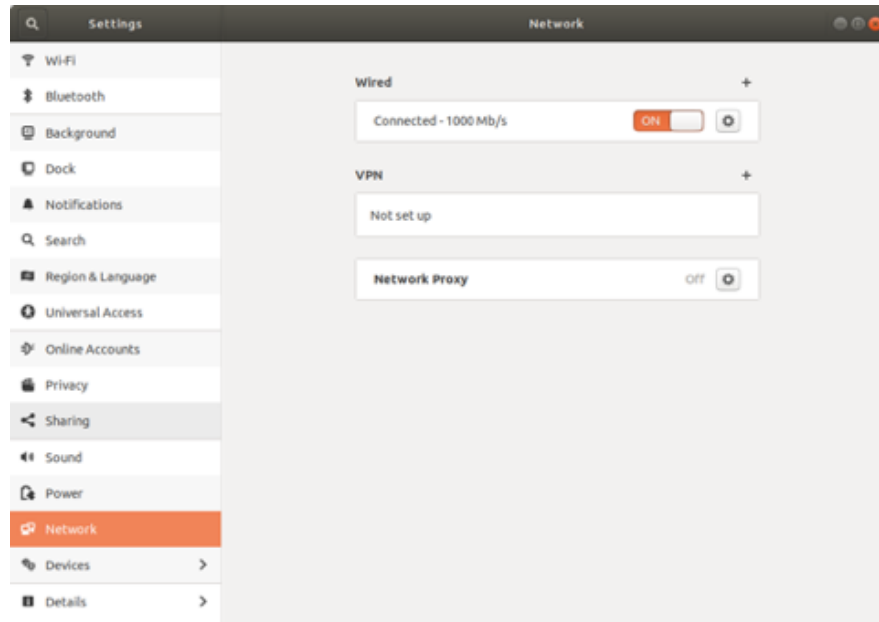
In terminal type "*sudo dpkg-reconfigure tzdata*", you will be asked for "Administrator" user password.

In text window select as Geographic area None of the above, then select UTC.

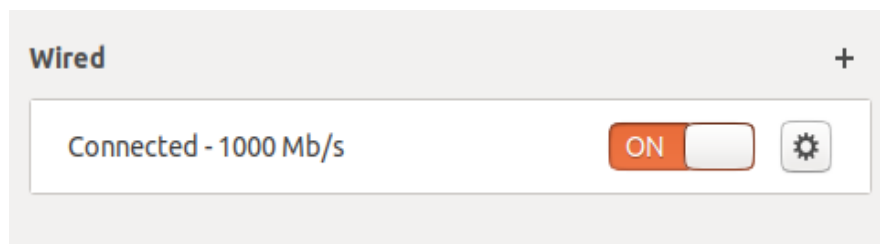
2.2.2 Network settings

Log in as any user. Click on Activities on left upper desktop area and find settings application. Open “Settings” window by clicking on  icon:

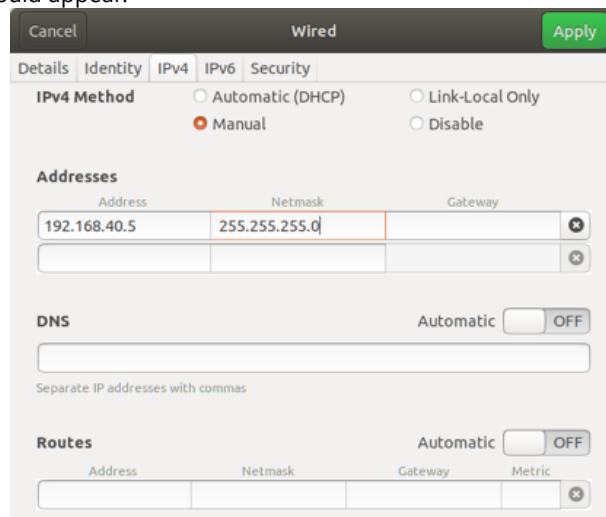
After clicking “Settings” window should appear. Select Network item.



Then click on last button in Wired area.



“Wired” window should appear.



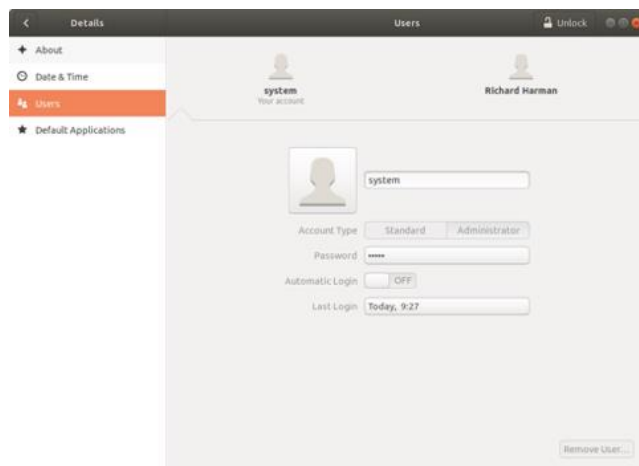
Select IPv4 Settings tab and Manual Method. In field Addresses fill correct IP address, netmask and gateway for this workstation (according IP Configuration). DNS and Routes fields set to OFF.

After filling all fields press Apply button.

Note: If you are not logged as "Administrator" user, system will ask for "Administrator" user password.

2.2.3 Managing users

Log in as any user. Open "Settings" window. Select Details and then Users item. "User accounts" window should appear.



2.2.3.1 Adding a new user account

- Click Unlock in the top right corner. System will ask for "Administrator" user password.
- In the list of accounts on the left, click the + button to add a new user account.
- If you want the new user to be "Administrative" user, select Administrator for the account type. For application user is recommended to set Standard account type. Enter the new user's full name. The username will be filled in automatically based on the full name. The default is probably OK, but you can change it if you like
- Click Add button.
- The account is initially disabled until you choose what to do about the user's password. Under Login Options click Account disabled next to Password. Select Set a password now from the Action drop-down list, and have the user type their password in the New password and Confirm password fields. (Button next to the New password field can be used to random generate secure password). These passwords are hard for others to guess, but they can be hard to remember, be careful.
- Click Change button.

2.2.3.2 Modifying a user account

- Click Unlock in the top right corner. System will ask for "Administrator" user password.
- Click on the name of a user in the list of users, then click on the text entry next to any of the following options:
 - Account type:
 - Language:
 - Password:
 - Automatic Login:
- You may also change the username by clicking on the username at the top and entering a new name.

2.2.3.3 Deleting a user account

- Click Unlock in the top right corner. System will ask for "Administrator" user password.
- Select a user from the list and click -.
- Ubuntu will deactivate the user's account, and you can choose whether to remove the user's home folder or leave it in place.

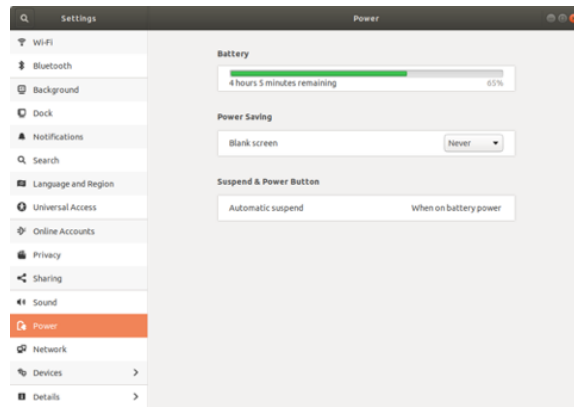
2.2.4 Lock screen options.

There are 2 different settings, one for blank screen and second for locking screen after blank.

2.2.4.1 Set timeout of “Blank Screen”

Open Settings. There select the Power settings tab. Now the Power option will be visible.

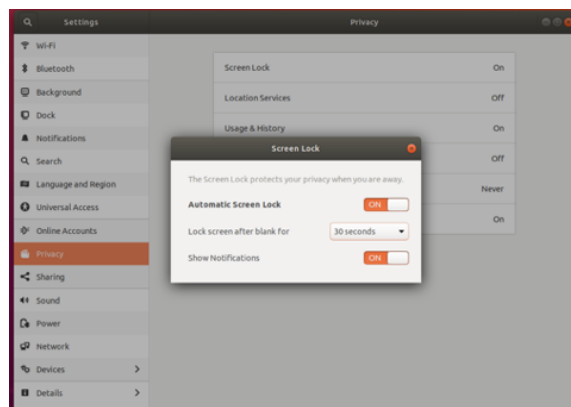
There you can set a time for blanking the screen.



2.2.4.2 Set timeout of “Lock screen after blank”

Open Settings. There select the Privacy settings tab. Now the Screen Lock option will be visible. When clicked, a modal window with the specific settings will open.

There you can toggle Automatic Screen Lock on/off, set a time for locking the screen or respectively binding it to the screen turning off via Lock screen after blank for, last but not least Show Notifications lets you decide if notifications should be shown on the lock screen.



2.2.5 Time synchronization with NTP

First of all, you should turn off timesyncd. Type in terminal:

```
$ sudo timedatectl set-ntp no
```

System will ask for "Administrator" user password.

Verify that timesyncd is off:

```
$ timedatectl
```

Look for systemd-timesyncd.service active: no in the output. This means timesyncd has been stopped.

Edit /etc/ntp.conf to add/remove server lines. You can use e.g. gedit text editor, (in example).

To edit file, type in terminal `sudo gedit /etc/ntp.conf`. By default these servers are configured:

```
# Use servers from the NTP Pool Project. Approved by Ubuntu Technical Board
# on 2011-02-08 (LP: #104525). See http://www.pool.ntp.org/join.html for
# more information.
server 0.ubuntu.pool.ntp.org
server 1.ubuntu.pool.ntp.org
server 2.ubuntu.pool.ntp.org
server 3.ubuntu.pool.ntp.org
```

Example of configuration:

- Server: 172.16.10.200
- Server: 172.16.10.201

After changing the config file you have to reload the ntp service, type in terminal:

```
sudo service ntp reload
```

System will ask for "Administrator" user password

Note: NTP configuration needs some extra system packages, so it will be accessible only after ifdps system installation, see next chapter.

To view NTP status, type in terminal:

```
sudo ntpq -p
```

2.3 IFDPS system installation

2.3.1 iFDPS system workstation types

There are two types of workstations in ifdps system, server workstation and client workstation. Client workstation contains ifdps client(lfsd) for working with FPL strips(see user documentation). Server workstation contains the same lfsd application but in addition also ifdps server(ifds) for FPL processing and interface to external system(s)(AFTN). Both lfsd clients should be connected to ifds server and have access to ifdps database, client workstation through NFS file system(after installation client workstation folder /mnt/nfs/home is mapped to server workstation folder /home). This means that there are also two types of installations, server installation and client installation.

Server installation ensures:

- Install all necessary system support and development packages(NFS support,database support,Qt development)
- Install special device drivers for touch monitor
- Configure NFS file system server
- Support for time synchronization(NTP)
- Install and configure ifdps server(ifds) software. This installation also ensures creating ifdps user for running server and installing ifdps database.
- Install and configure ifdps client(lfsd) software.

Client installation ensures:

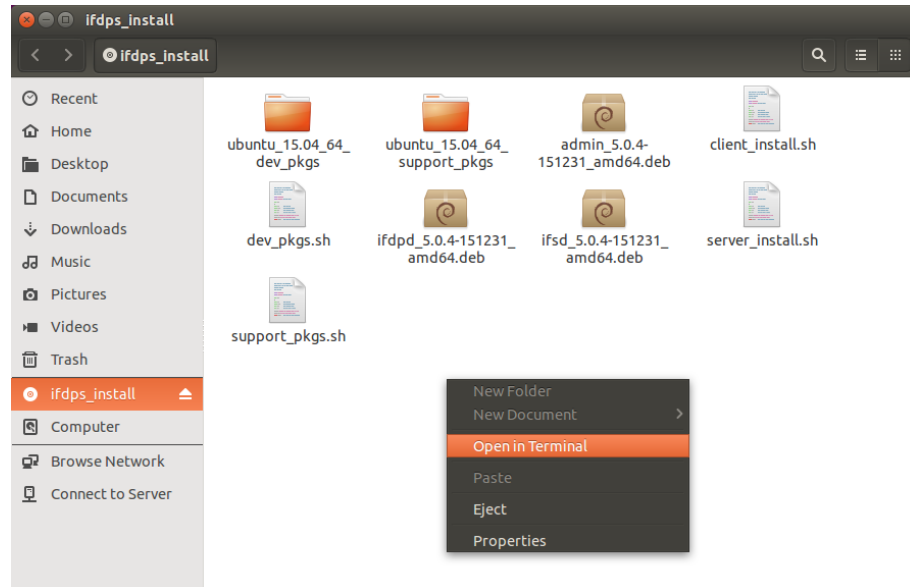
- Install all necessary system support and development packages
- Install special device drivers for touch monitors
- Configure NFS file system client
- Support for time synchronization(NTP)
- Install and configure ifdps client(lfsd) software.

2.3.2 iFDPS system initial installation

This installation should be applied after (re)installing operating system and post installation settings.

Log in as “Administrative” user. Put the ifdps system install DVD into the DVD-drive.

Window with content of installation DVD should appear.



Right-click on window background and select Open in Terminal.

For server installation type in terminal:

```
./server_install.sh
```

For client installation type in terminal:

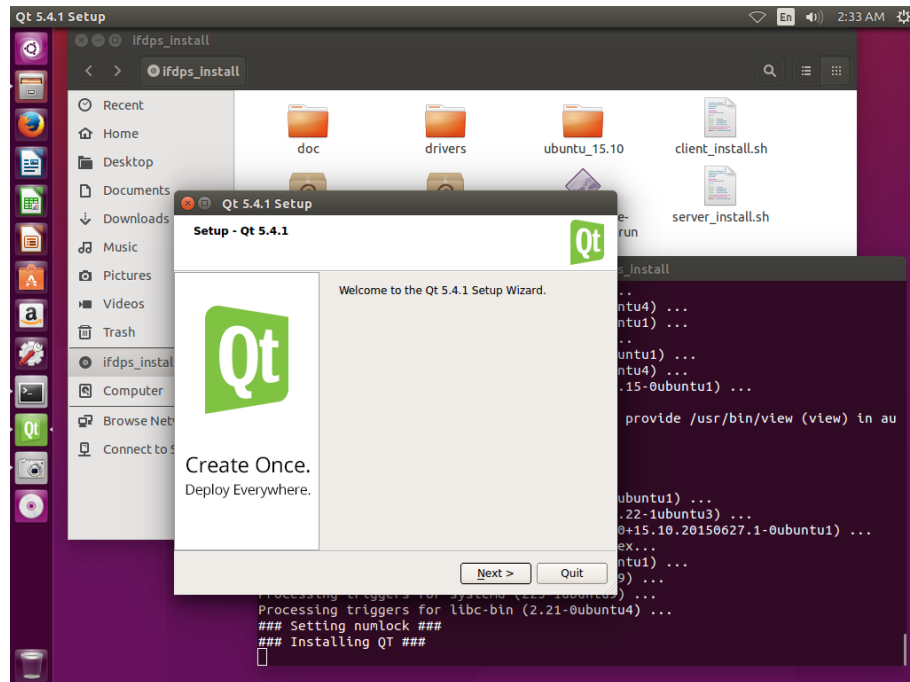
```
./client_install.sh
```

The installer will ask for “Administrator” user password.

Note: In case of client installation installer will also ask for server hostname (ifdps server) due to correct NFS client configuration. If you entered wrong value or server hostname will change the administrator must change hostname manually in /etc/hosts file, in line with ifdps_srv symbolic name. To check this settings, type in terminal ping ifdps-srv and you should get respond from ifdps server.

```
administrator@ifdps-server: /media/administrator/ifdps_install
administrator@ifdps-server:/media/administrator/ifdps_install$ ./server_install.sh
[sudo] password for administrator:
Selecting previously unselected package libnfsidmap2:amd64.
(Reading database ... 170930 files and directories currently installed.)
Preparing to unpack .../libnfsidmap2_0.25-5_amd64.deb ...
Unpacking libnfsidmap2:amd64 (0.25-5) ...
Preparing to unpack .../libsqlite3-0_3.8.7.4-1ubuntu0.1_amd64.deb ...
Unpacking libsqlite3-0:amd64 (3.8.7.4-1ubuntu0.1) over (3.8.7.4-1) ...
Selecting previously unselected package libssh2-1:amd64.
Preparing to unpack .../libssh2-1_1.4.3-4.1_amd64.deb ...
Unpacking libssh2-1:amd64 (1.4.3-4.1) ...
Selecting previously unselected package libtirpc1:amd64.
Preparing to unpack .../libtirpc1_0.2.5-1_amd64.deb ...
Unpacking libtirpc1:amd64 (0.2.5-1) ...
Selecting previously unselected package mc-data.
Preparing to unpack .../mc-data_3%3a4.8.13-3_all.deb ...
Unpacking mc-data (3:4.8.13-3) ...
Selecting previously unselected package mc.
Preparing to unpack .../mc_3%3a4.8.13-3_amd64.deb ...
Unpacking mc (3:4.8.13-3) ...
Selecting previously unselected package nfs-common.
Preparing to unpack .../nfs-common_1%3a1.2.8-9ubuntu8_amd64.deb ...
Unpacking nfs-common (1:1.2.8-9ubuntu8) ...
Selecting previously unselected package nfs-kernel-server.
```

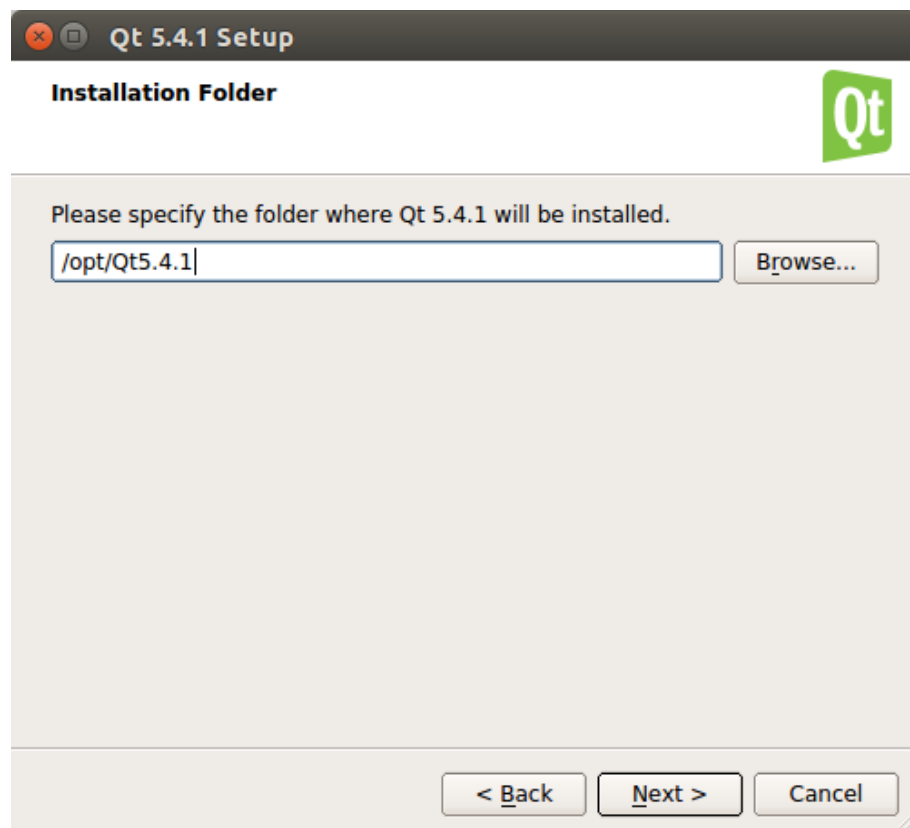
Part of server and client installation is QT installation. This is in graphics mode and after finishing installer will return again to terminal mode to finish all installation.



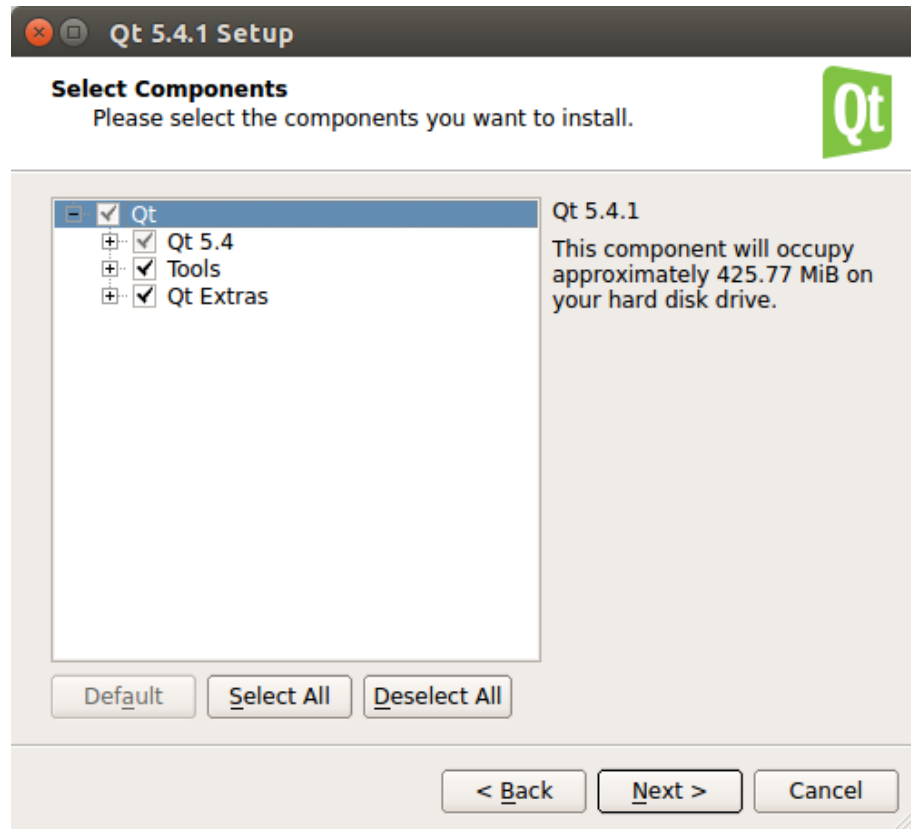
QT installation starts with the next window.

Note: In case of QT 5.12.2, installation starts with two additional windows, just click next on first and skip on second window. Then the installation continues in the same way.

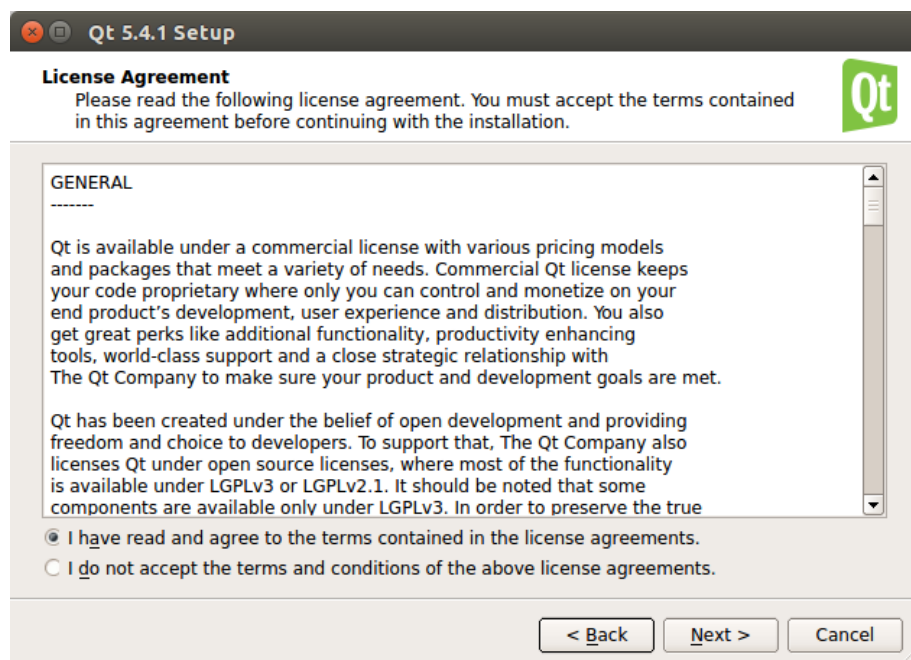
Click Next button to start QT installation. Next window should appear.



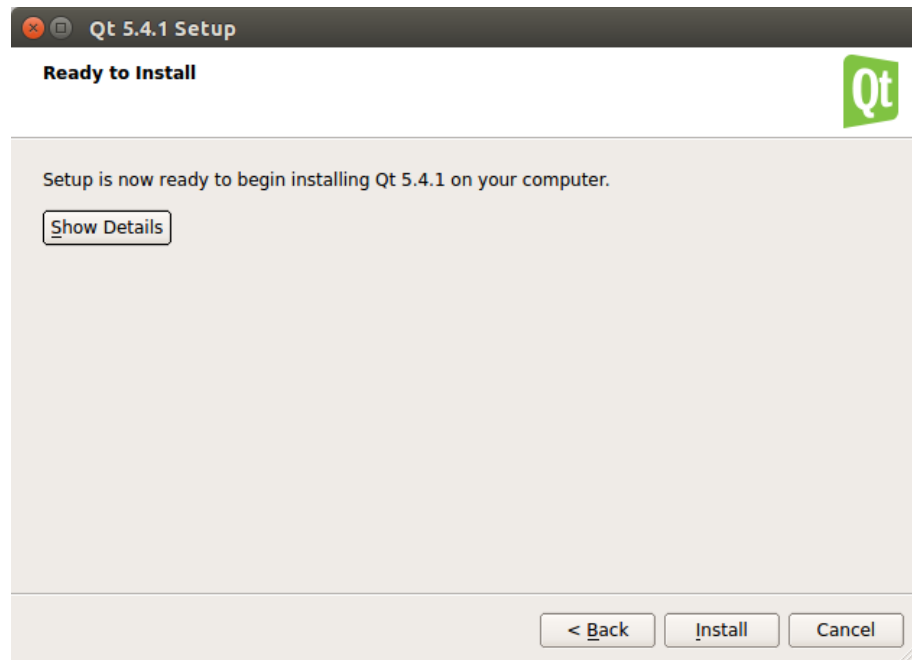
Let installation folder for QT with default value and click Next button to continue installation. Next window should appear:



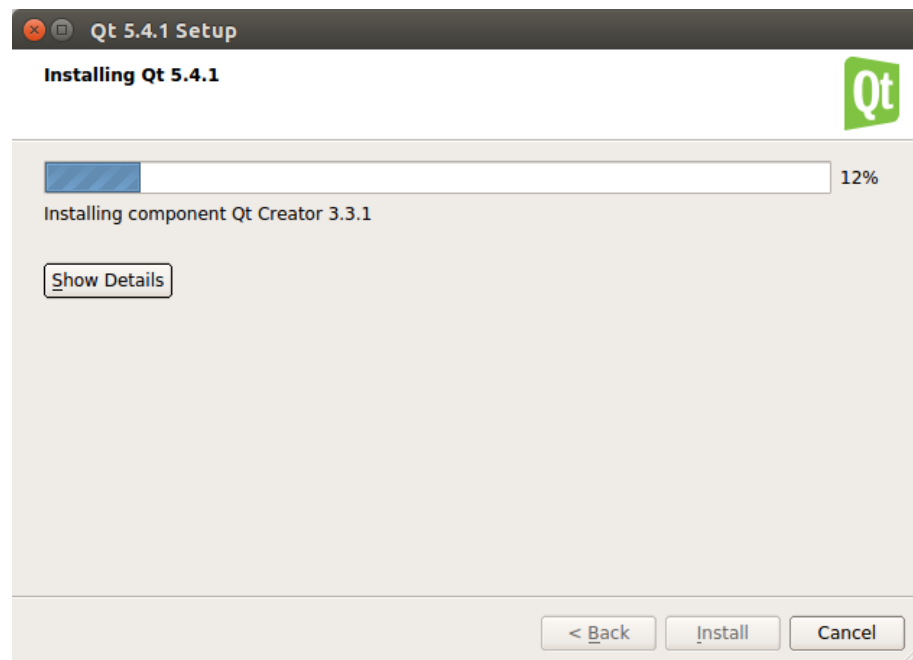
Let all default options checked and press Next button to continue. Next window should appear:



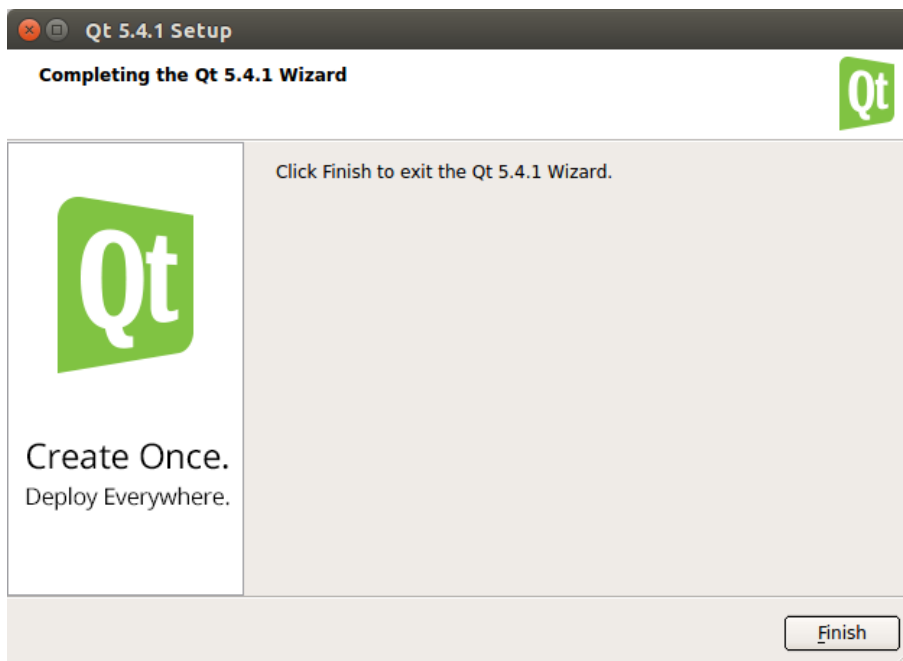
Check agree to the terms options and press Next button to continue installation. Next window should appear.



Press Install button to continue installation. Window with installation progress should appear.



After finishing QT installation next window should appear.



Press Finish button and installer will continue in terminal mode.

After finishing script ifdps system installation is done.

Close the terminal, eject DVD medium and restart computer. After restarting computer ifdps server(ifds) will run automatically in context of ifdps user with default configuration, for running ifdps client application(lfscd) you should login to the system and run it manually or set auto-run after login for selected user. Application will run with the default configuration.

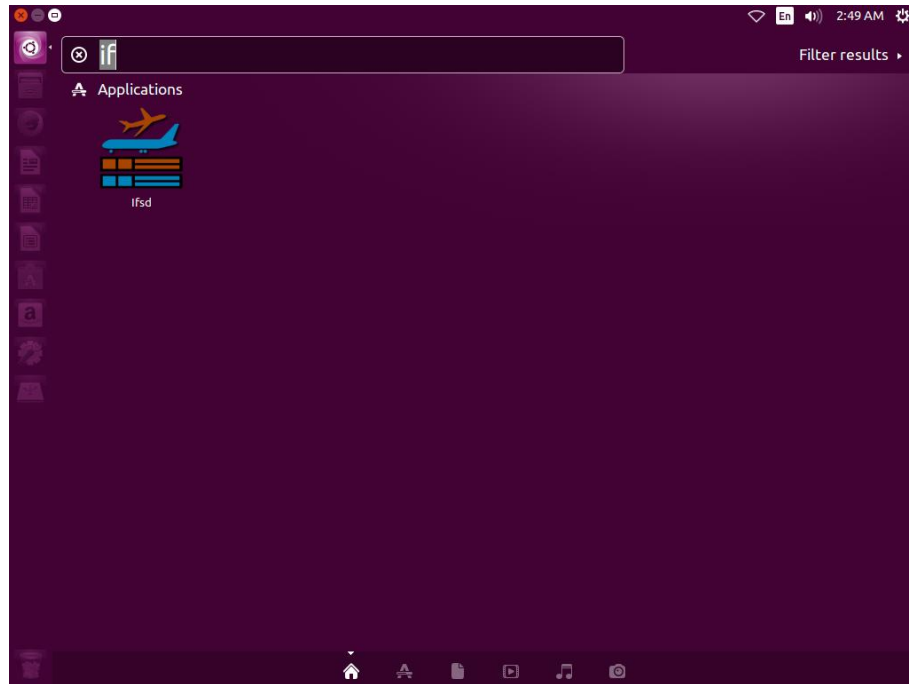
Note: Due to installation USBHID correction package as part of ifdps installation, please, plug-in the USB cable from Wacom monitor to 2nd USB port from the bottom on back side of the PC. See in a picture. The USB cable is marked as "USB monitor" .



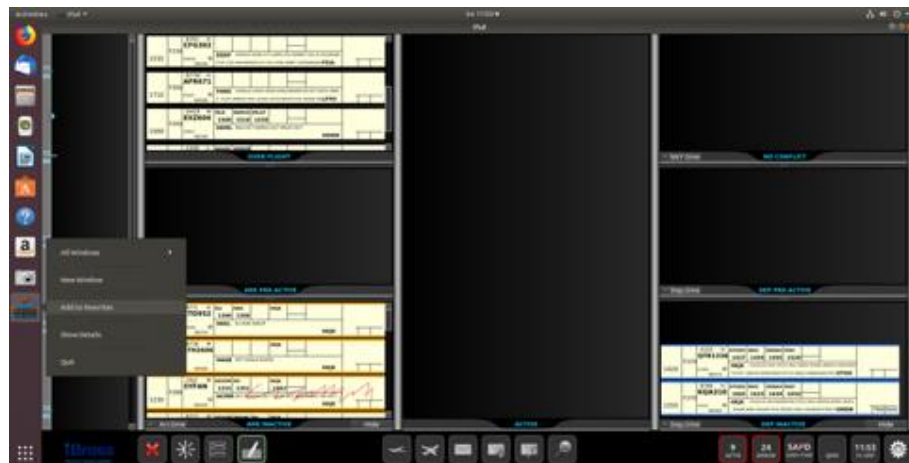
2.3.3 Running Ifsd application manually

Log in as any user. Click on Activities on left upper desktop area

Start typing application name (Ifsd), until you see Ifsd icon in Applications section of the dash.

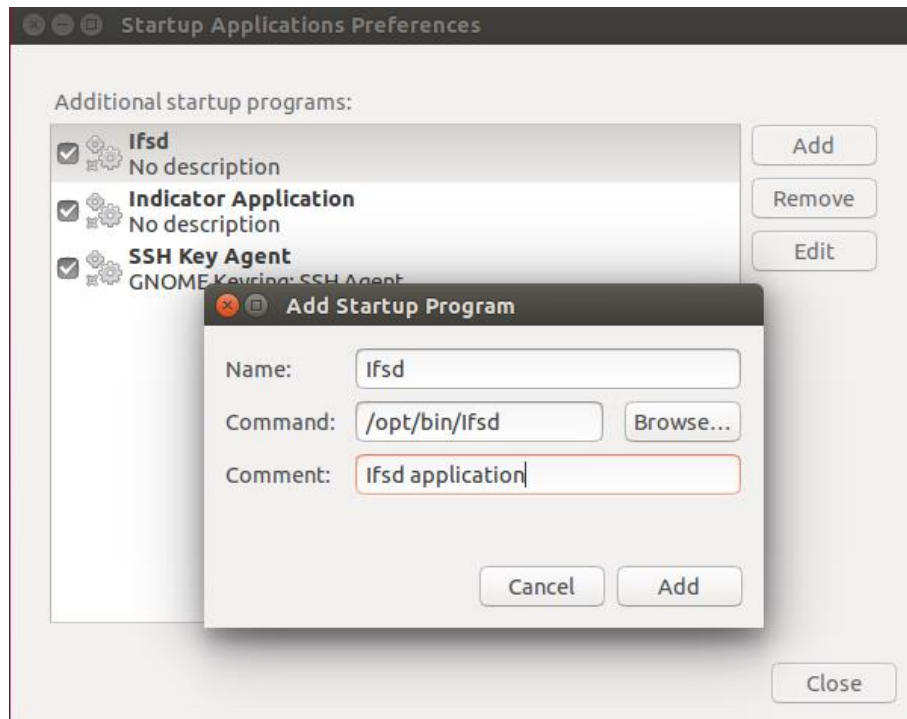


Click on the Ifsd icon. Application should start. If you want to have an icon for Ifsd start on the desktop, right click on application icon and select Add to Favorites. From now you can start application also from here. To remove icon right-click on application icon and select Remove from Favorites.



2.3.4 Running Ifsd application automatically after user login

Click on Activities on left upper desktop area. Start typing startup applications. Click on the icon Startup Applications. Startup Applications Preferences window will appear. Press Add button to add Ifsd application. Fill appropriate text fields according picture below. Then press add button and close Startup Applications Preferences window. Upon next log-in application Ifsd will start automatically.



3. UPGRADE AND CONFIGURATION

3.1 Upgrade of operational system

Latest version of operation system is provided.

Updates that are proven and recommended will be provided by supplier of the system.

3.2 Upgrade of application software

This installation should be applied for installing standalone ifdps applications (ifds or Ifsd), for example, in case of existing newer version with some improvements (error corrections, enhanced functionality,...).

Log in as "Administrative" user.

Open terminal.

For installing Ifsd client type:

```
sudo dpkg --install ifsd*.deb
```

For installing ifds server type:

```
sudo dpkg --install ifds*.deb
```

```
sudo /etc/init.d/ifds restart
```

Ifsd*.dep, ifds*.deb files are application packages and should be delivered to customer for ifdps system upgrade.

Installation of the server ("strips1" cpu):

Copy *server.deb* package to *Desktop*,
where *server.deb* = "ifds_10.0.1-160305_amd64.deb"

Then type:

```
: cd /Desktop
: sudo --install server .deb
// unpack updated server software.
// be careful to type proper name of the package
: sudo /etc/init.d/ifds stop
// stop the server
: sudo /etc/init.d/ifds start
// restart the server
```

Installation of the client (both "strips1" and "strips2" cpu-s):

Copy *client.deb*, package to the Desktop,
where *client.deb* = "ifsd_10.0.1-160305_amd64.deb"

Then type:

```
: cd /Desktop
: sudo dpkg --install client.deb
```

Stop application using *Exit* button

Restart application using icon in launch dock

3.3 iFDPS configuration

3.3.1 iFDPS database

System ifdps using SQLITE database. Each logical database is stored in one file. Database for ifdps system (ifdps database) is located in ifdps.db file. The owner of this file is ifdps user created automatically during server (ifds) installation. This file resides in ifdps user home subdirectory (by default in /home/ifdps/.config/ifdps/ifds/ifdps.db), where , it depends on ifds server configuration file, see below. It is created automatically (copied from initial one from installation) when ifds server is running for the first time. If you want to see or modify this database using SQL language you must to connect to this database(file), in database console (sqlite3). To connect, type in terminal: sqlite3 /home/ifdps/.config/ifdps/ifds/ifdps.db, or other appropriate file path for database file.

Note: If you want to modify ifdps database you must have right to modify db-file. To ensure this you should be log in as ifdps user or run sqlite3 using sudo command.

To log as ifdps user type in terminal in server workstation:

```
sudo su ifdps
```

```
sqlite3....(login_to_db)
```

To run sqlite3 using sudo type in terminal in server workstation:

```
sudo sqlite3....(login_to_db)
```

3.3.2 iFDPS system parameters

System parameters are stored in ifdps database, strictly speaking in config table. If a change of some system parameter is required, editing the config table must take place. Table config contains of 2 columns, name and value. Column name is the name of the parameter, column value represents the value of this parameter. If the given parameter is not in the config table, the default value of the parameter is used. Default parameter value is encoded in applications. Not all system parameters has default values.

Examples of working with ifdps system parameters:

- **Log to ifdps database in sqlite console.**
- **Insert system parameter 'del_after_landing' with value 12.**
- **Change of system parameter 'del_after_landing' with value 6.**
- **Print of system parameter 'del_after_landing'.**

```
administrator@ifdps-server: ~
administrator@ifdps-server:~$ sudo sqlite3 /home/ifdps/.config/ifdps/ifdpd/ifdps.db
SQLite version 3.8.7.4 2014-12-09 01:34:36
Enter ".help" for usage hints.
sqlite> insert into config(name,value) values('del_after_landing','12');
sqlite> update config set value=6 where name='del_after_landing';
sqlite> select * from config where name='del_after_landing';
35|del_after_landing|6
sqlite>
```

After all necessary changes of system parameters the ifdpd server must be restarted to take effect!

To do it, type in terminal in server workstation (log in as "Administrator" user)

```
sudo /etc/init.d/ifds restart
```

System will ask for "Administrator" user password. All system parameters and their values are described below. Predefined value means initial value stored in ifdps database. NULL means that this parameter has no defined value.

known_aftn_originators – list of known AFTN sender addresses separated by spaces , for automatic processing, otherwise not processed(appear in error queue), special value '*****' is for accepting from any sender address.

Default value: NULL

Predefined value: '*****'

fdp_aftn_address – list of input AFTN addresses for ifdps system, input AFTN messages with different destination addresses are ignored.

Default value: NULL

Predefined value: '*****'

fdp_out_aftn_address – output AFTN address for ifdps system. Source address for all output AFTN messages

Default value: NULL

Predefined value: '*****'

del_after_est_arr – How long after ETA in minutes FPL will be removed from system.

Default value: 120 min

Predefined value: NULL

del_after_real_arr – How long after landing in minutes(message ARR,or manual) FPL will be removed from the system.

Default value: 60 min

Predefined value: 10 min

del_dep_after_takeoff – How long after takeoff in minutes(message DEP, or manual) FPL will be removed from the system. Apply only for DEPARTURES. Value -1 means that criterium does not apply at all.

Default value: 30 min

Predefined value: NULL

eobt_dep_interval – In which range from EOBT- value to EOBT+ value is able to apply takeoff or DEP message.

Default value: 15 min

Predefined value: 300 min

del_err_msg_interval – How long after creating in hours ,message in error queue will be removed.

Default value: 24 hod

Predefined value: 3 hod

update_fpl_by_duplicate_fpl – If set up ('Y') system will replace old FPL by new one without error indication(duplicate FPL).

Default value: No

Predefined value:Yes

airport - ICAO code for airport for which is system intended.

Default value: NULL

Predefined value: HKNW

fir – ICAO code for FIR for which is system intended.

Default value: NULL

Predefined value: HKNA

dep_before_eobt – How long before EOBT in minutes FPL appears in section DEPARTURE. Apply only for DEPARTURES or LOCAL flights.

Default value: 60 min

Predefined value: NULL

cleared_before_eobt – How long before EOBT in minutes FPL appears in section CLEARED. Apply only for DEPARTURES or LOCAL flights.

Default value: 30 min

Predefined value: NULL

arr_before_eta – How long before ETA in minutes FPL appears in section ARRIVAL. Apply only for ARRIVALS or LOCAL flights.

Default value: 40 min

Predefined value: NULL

db_delete_older_then - Delete older records than this value in days, see chapter Database management

Default value: 36

Predefined value: 100

check_dep_before_arr - If set up ('Y') , system after receiving ARR message or manual landing check if was detected takeOff (DEP message or manually)

Default value: No

Predefined value: No

del_fpl_after_eobt – How long after EOBT in minutes FPL will be removed from system. This value should be big enough. Intended for mini-fpl with EOBT only.

Default value: 24 hod

Predefined value: NULL

3.3.3 iFDPS server configuration file

Configuration file name is ifdpd.conf. It is located in ifdps user(created during ifdpd installation) home subdirectory ~/.config/ifdps/ifdpd directory. It consist of several sections. Each section started with its name enclosed in brackets e.g. [aftn_tcp] and contains configuration for specific function. File is created automatically when process ifdpd is running for the first time,with default values. If some default values are not advisable you can change it. After changing some value you need to restart ifdpd server. To do it, log in as "Administrator" user,open the terminal and type

sudo /etc/init.d/ifdpd restart

System will ask for "Administrator" user password. Description of configuration file is following.

[aftn_tcp]

Configuration section for setting the connection to AFTN over TCP/IP

use = [true | false]

Flag to indicate whether or not to use this function. Example use=true

hosts = [host1,...]

List of TCP/IP address(es) separated by comma to connect to AFTN TCP/IP server(AFTN switch). Example: hosts = 172.16.10.3

ports = [port_number1,...]

List of port number(s) to separated by comma to connect to AFTN TCP/IP server. One port is per AFTN channel. Example ports=11125

local_host = [| IP]

The local used TCP address for connection request. This address appears like remote address for AFTN TCP server. If it is configured on server side, AFTN TCP server reject connection request from other addresses. Address can be empty, in that case **local_host** and **local_port** parameters are not bind to connection request.

local_port = [port_number]

The locally used TCP port for connection request. If port number is 0 a random port is chosen. This port appears like remote port for AFTN TCP server. If it is configured on server side, AFTN TCP server reject connection request from other ports.

out_channels = [channel1,channel2,...]

Identification codes for output channels. It may contain only 3 upper-case characters.

Example: out_channels=ALA

in_channel = [channel1,channel2,...]

Identification codes for input channels. The code may contain only 3 upper-case characters.

Example: in_channels=LAL

ctrl_addresses = [adr1,adr2,...]

AFTN address for sending AFTN service messages (e.g. a MIS report on missing telegrams).

Example: ctrl_addresses=UKKKYXYA

frame_indicator_length = [1,2,4]

Size of message header in bytes. It gives the size of subsequent TCP frame with AFTN message.

Example: frame_indicator_length=2

frame_indicator_endian_type=[BigEndian | LittleEndian]

The order in which the bytes are stored in the header.

Example: frame_indicator_endian_type=BigEndian

is_server = [true | false]

If ifdpd is TCP client or server for AFTN_TCP communication.

Example: is_server = false

[database]

Configuration section for setting the connection to the database

database_name = name_of_database

Name of the logical database, or name of the file(in case of QSQLITE database_type), which contains ifdps database tables

Example: database_name=/home/ifdps/.config/ifdps/ifdps.db

database_type=[QSQLITE|QMYSQL|...]

Type of the database

Example: database_type=QSQLITE

hostname = ip_address

IP adress/hostname, on which the database is running

Example: hostname=127.0.0.1

password = password_string

Password to gain access to the ifdps database tables

Example: password=ifdps

username = db_user_name

Database user with access to ifdps database tables

Example: username=ifdps

Note: For supported QSQLITE database type password,username and hostname settings has no effect.

[log]

Configuration section for logging settings.

use=[true|false]

Flag to indicate whether or not to use logging

Example: use=true

path = path_to_log_directory

Path to log directory

path=/home/ifdps/.config/ifdps/ifdpd/Logs

days_to_delete = number_of_days

Specifies the retention period of old log files in days.

Example: days_to_delete=10

max_file_size = file_size

Gives maximum size of the configuration file in MB. On exceeding this size a new log file is created.

Example: max_file_size=10

[server]

Configuration section for server settings: (for FDO client connections)

port_number = port

Port number for communication with ifdpd FDO clients.

Example: port_number=5099

HB = hb_timeout

Period for sending Heartbeat signal [sec]. If the receiving station doesnt receive Heatbeat for interval time*3 it is pronounced as non-living.

Example: HB=5

[xml-server]

Configuration section for server settings: (for RDD client xml connections)

port_number = port

Port number for communication with ifdpd RDD clients.

Example: port_number=5102

HB = hb_timeout

Period for sending Heartbeat signal [sec]. If the receiving station doesnt receive Heatbeat for interval time*3 it is pronounced as non-living.

Example: HB=5

[StripPrinter]

Configuration section for strip printer settings

ip = ip_address

IP address/hostname of the strip printer.

Example: ip = 192.168.40.132

port = port_number

Port number to connect to strip printer.

Example: port = 9100

use = [true | false]

Flag to indicate whether or not to use strip printer

Example use=true

3.3.4 Default configuration

3.3.4.1 IP Configuration

Wilson / server / 192.168.40.130 / CZC51022BX

Wilson / client / 192.168.40.131 / CZC51022BV

Wilson / printer / 192.168.40.132 / 322609

3.3.4.2 Passwords

Fill here your passwords:

☐ Server root:

☐ Client root:

4. TROUBLESHOOTING

4.1 Log files

4.1.1 iFDPS server (ifdpd) logging

All important events in server processing are writing to log file. Log settings and properties are described in previous chapter(configuration, [log] section). Owner of log file is ifdps user. One log message consist of date and time of event, category of log message and description of event. There are five categories of log messages: Debug,Info,Warning,Critical,Fatal.

Debug – Debug message, print only if application is running with -d switch.

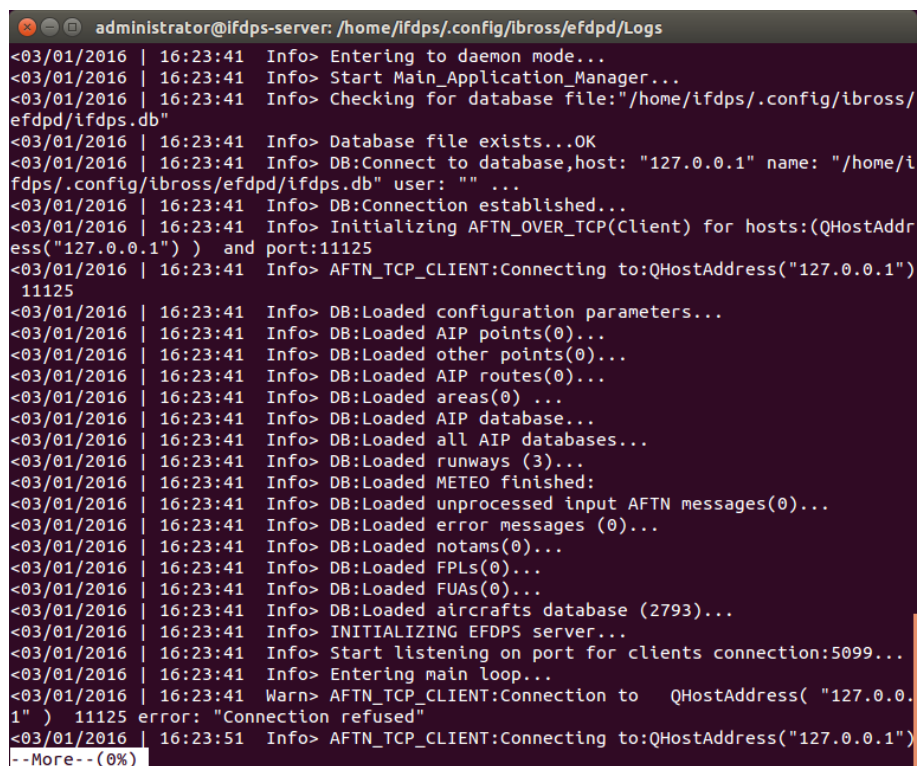
Info – Informative message.

Warning – Message indicates not serious error, application can continue.

Critical – Message indicates serious error, but application can continue.

Fatal – Message indicates fatal, non-recoverable error, application cannot continue, after printing this message application finish.

Example of ifds log file:



```
administrator@ifdps-server: /home/ifdps/.config/ibross/efdpd/Logs
<03/01/2016 | 16:23:41 Info> Entering to daemon mode...
<03/01/2016 | 16:23:41 Info> Start Main_Application_Manager...
<03/01/2016 | 16:23:41 Info> Checking for database file: "/home/ifdps/.config/ibross/efdpd/ifdps.db"
<03/01/2016 | 16:23:41 Info> Database file exists...OK
<03/01/2016 | 16:23:41 Info> DB:Connect to database,host: "127.0.0.1" name: "/home/ifdps/.config/ibross/efdpd/ifdps.db" user: "" ...
<03/01/2016 | 16:23:41 Info> DB:Connection established...
<03/01/2016 | 16:23:41 Info> Initializing AFTN_OVER_TCP(Client) for hosts:(QHostAddress("127.0.0.1") ) and port:11125
<03/01/2016 | 16:23:41 Info> AFTN_TCP_CLIENT:Connecting to:QHostAddress("127.0.0.1") 11125
<03/01/2016 | 16:23:41 Info> DB:Loaded configuration parameters...
<03/01/2016 | 16:23:41 Info> DB:Loaded AIP points(0)...
<03/01/2016 | 16:23:41 Info> DB:Loaded other points(0)...
<03/01/2016 | 16:23:41 Info> DB:Loaded AIP routes(0)...
<03/01/2016 | 16:23:41 Info> DB:Loaded areas(0) ...
<03/01/2016 | 16:23:41 Info> DB:Loaded AIP database...
<03/01/2016 | 16:23:41 Info> DB:Loaded all AIP databases...
<03/01/2016 | 16:23:41 Info> DB:Loaded runways (3)...
<03/01/2016 | 16:23:41 Info> DB:Loaded METEO finished:
<03/01/2016 | 16:23:41 Info> DB:Loaded unprocessed input AFTN messages(0)...
<03/01/2016 | 16:23:41 Info> DB:Loaded error messages (0)...
<03/01/2016 | 16:23:41 Info> DB:Loaded notams(0)...
<03/01/2016 | 16:23:41 Info> DB:Loaded FPLs(0)...
<03/01/2016 | 16:23:41 Info> DB:Loaded FUAs(0)...
<03/01/2016 | 16:23:41 Info> DB:Loaded aircrafts database (2793)...
<03/01/2016 | 16:23:41 Info> INITIALIZING EFDPS server...
<03/01/2016 | 16:23:41 Info> Start listening on port for clients connection:5099...
<03/01/2016 | 16:23:41 Info> Entering main loop...
<03/01/2016 | 16:23:41 Warn> AFTN_TCP_CLIENT:Connection to QHostAddress( "127.0.0.1" ) 11125 error: "Connection refused"
<03/01/2016 | 16:23:51 Info> AFTN_TCP_CLIENT:Connecting to:QHostAddress("127.0.0.1")
--More-- (0%)
```

4.1.2 iFDPS client (lfdd) configuration

Configuration of client is described in lfdd user documentation.

4.2 Supporting scripts

4.2.1 Backup scripts

4.2.1.1 Server_log script

This script backups all current server (ifds) configuration, ifdps database and log files.

To run it, just open the terminal and type

: server_log.sh

The script will ask for "Administrator" type password.

Resulting file with backup should appears in Desktop and looks like i.e.:

: kumasi1-05:06:27-server.tgz,

where kumasi1 is hostname followed by time of creation.

4.2.1.2 Client_log script

This script backups all current client (lfds) configuration and log files in actual logged user.

To run it, just open the terminal and type

: client_log.sh

Resulting file with backup should appears in Desktop and looks like

: kumasi2-05:10:12-client.tgz

where kumasi2 is hostname followed by time of creation.

In case of reporting problem, please also send these files.

4.3 Troubleshooting

Strip printer does not work

Check if strip printer is accessible on network, typing "**ping strip_printer_ip**". If no you have to check printer connection or printer settings according printer documentation. If yes check ifds server configuration file, section [StripPrinter] for correct configuration.

No connection to server (client)

Check if ifds server is running typing `pgrep ifds` in terminal. If no proceed to 4.3. If yes, check Ifsd connection settings according User documentation.

No connection to database (client)

Check if the database file **ifdps.db** exists in appropriate file path according database settings in Ifsd application.

Server does not running

Try to restart the computer or run ifds manually by typing in terminal "**sudo /etc/init.d/ifds start**" command.

Check if still running with "**pgrep ifds**" command. If not check the log file for reason.

Client workstation does not turn off

If you try turn off your client workstation after turning off server workstation, this turning off will last very long time due to NFS configuration and timeouts. Wait for appropriate time or turn on server workstation again and turn off again only after client workstation is switch off.

Screen become frozen

Press "**Ctrl+Alt+F1**" to switch to one of the virtual text consoles. Log in as "Administrator" type and reboot system by typing reboot command.

If this doesn't help try these steps:

Hold down "**Alt+SysRq (PrtScnKey)**" keys combination. There will be no visible output, while hold down the key combination, then press these keys in the order, one by one, "**REISUB**" and the system will be reboot after B is typed in the safe way.

If all previous attempts fails, as the last option push the power button for a while to switch off the computer. Then switch on computer again.

4.4 Database maintenance.

4.4.1 Script deleteArchives.sh

Script do the following:

- Stop the ifds server if it is running
- Delete older records than is set by system parameter "db_delete_older_then" (flight plans, messages, operators loggings)...
- Optimize database (reduce database size and increase database speed)
- Starting ifds server if it was running before.

```
administrator@ifdps-server:~$ deleteArchives.sh
```

```
Stopping ifds process...
```

```
[sudo] password for administrator:
```

```
Deleting ifdps archives ...
```

```
archive period
```

```
100
```

```
Deleting output messages...
```

```
Deleting flight plans...
```

```
Deleting input messages...
```

```
Deleting client connections...
```

```
Rebuilds the database file...
```

```
Finishing delete script...
```

```
Starting ifds process...
```

In this case we are deleting records older than 100 days.

Note: It is recommended that you run this script periodically e.g. every day. If you run script after longer period of time its running time may be longer. Server should not run during script is working.

4.5 Support of the client

For support please contact following address:

sw@ibross.aero

or visit

www.ibross.aero

END OF DOCUMENT

NOTES
